

# COM-HPC Server: A New Standard For Modular Edge Servers

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The PCI Industrial Computer Manufacturers Group (PICMG) will soon officially adopt a computer-on-module for high-performance computing (COM-HPC) standard. A part of the new standard is the COM-HPC server specification. It was developed for high-end edge server computing, which eagerly awaits new high-speed interfaces such as PCI Express 4.0 and 5.0 as well as 25 gigabit Ethernet. Compared to established COM Express Type 7-based server-on-modules, the new specification has almost twice the interface bandwidth with its 800 signal lines and therefore provides a significantly greater range of functions besides the performance boost.

The COM-HPC server addresses the entire range of edge servers as the new server-on-modules has a maximum power consumption of 300 watts and can therefore use the powerful Intel Xeon and AMD EPYC server processors. However, the first modules are not expected to become available until the launch of the next Intel embedded server processors.

COM-HPC modules can currently support up to one terabyte RAM on up to eight SO-DIMM sockets. The memory hosted on the modules cannot be made directly available to other parallel processors via Intel Ultra Path Interconnect (UPI) or AMD

Infinity Fabric Inter socket. Instead, it is exclusively available to each individual COM-HPC module. Dual-die/single-socket Cascade Lake processor technology with up to twelve-socket RAM is also not feasible with COM-HPC.

## Master/slave function for artificial intelligence

The roadmap nevertheless includes multi-module installations on carrier boards since COM-HPC allows master/slave configurations. However, this feature is not so much designed for dual-processor carrier boards, where two or more COM-HPC server modules could be operated in master/slave mode. Rather it aims to enable communication between COM-HPC server modules and COM-HPC FPGA and/or COM-HPC GPGPU modules—a third application area of the COM-HPC standard besides COM-HPC server and COM-HPC Client modules.

The master/slave function, available for the first time with computer-on-modules, was primarily implemented to make flat GPGPU or FPGA modules available in a form factor that is significantly more powerful than MXM. Hence, the COM-HPC standard also stands for the integration of functions such as artificial intelligence,

deep learning, and digital signal processing (DSP), which is important for telecommunications. With COM-HPC, all these functions can be made available in a robust, space-saving, and scalable manner.



Fig. 1: COM-HPC server designs enable load balancing at the edge not only through virtualisation